VoluSmoke V 1.0

VertexField

Table of Contents

Please start by watching the Quick Start Tutorial to get up and running in just a few minutes. If you want a deeper understanding of all features and workflows, the Advanced Tutorial provides a full walkthrough of the entire package.

Quick Starting Tutorial

Advance Tutorial

Overview

VoluSmoke is a production-ready, **volumetric-style smoke and fog toolkit for Unity**, built for real-time use across PC, console, and XR projects. It enables rapid creation of cinematic smoke plumes and atmospheric fog with minimal setup.

Instead of true volumetrics, VoluSmoke uses a **6-way lighting impostor shader combined with layered fog cards** to produce convincing depth and lighting at a fraction of the performance cost. This technique works best when the camera stays above or within the fog volume — making it an excellent fit for **isometric, top-down, and fixed-camera games**, while still suitable for **third-person and first-person** titles (see *Limitations* for important details).

The package provides **modular prefabs**, **Shader Graph–based materials**, and a simple, artist-friendly workflow that allows level designers and technical artists to drop in fog volumes and customize them quickly.

Key Features

Layered faux-volumetric technique that stacks lit transparent planes to create the illusion of depth, volume, and parallax—without the cost of true volumetrics.

6-way lighting shader for convincing smoke and fog shading that reacts to scene lighting, with support for shadows, lightprobes or APV

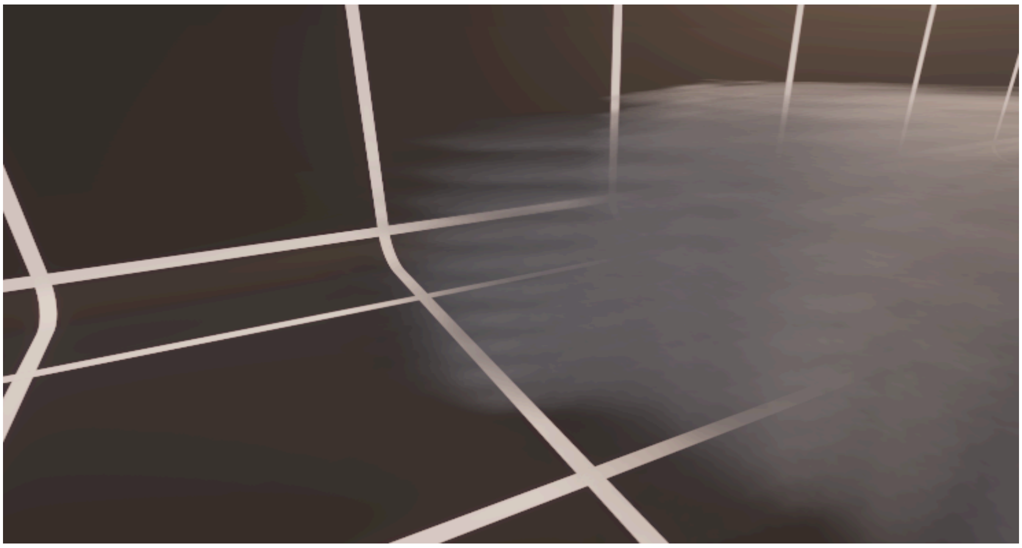
Non-intrusive and pipeline-friendly — built as a simple transparent lit shader and prefab workflow, making it compatible with most Unity projects, render pipelines, and 3rd-party tools.

Plug-and-play preset library with ready-to-use setups for common scenarios such as industrial vents, battlefield smoke plumes, isometric ground fog, magical mists, stylized ambience, and more.

Minimal setup workflow, enabling artists and designers to drop in a prefab and tweak a few parameters to achieve production-ready smoke and fog quickly.

Limitations

VoluSmoke uses stacked transparent layers to simulate volume. While this is significantly lighter than true volumetrics, **each added layer increases overdraw**, so keeping the layer count low is important for performance—especially on lower-end or XR hardware. Because the effect is created with flat slices, it cannot produce fully volumetric results when the camera moves through the smoke. At shallow viewing angles (e.g., looking sideways through the stack), the individual layers may become visible, particularly in first-person or third-person games. **Isometric or orthographic cameras hide this limitation extremely well**, making the technique ideal for low-lying ground fog, ambient mist, and top-down or fixed-camera titles.



Technical Requirements

- **Unity Version:**
  Unity 6 or newer.

- **Supported Render Pipelines:**
  • Universal Render Pipeline (**URP 12+**)
  • High Definition Render Pipeline (**HDRP 12+**)

- **Supported Platforms:**
  Windows, macOS, Linux, PlayStation 5, Xbox Series X|S, Meta Quest 2/3 (URP recommended), and SteamVR-compatible headsets.

- **Dependencies:**
  URP or HDRP variants require the corresponding pipeline package to be installed and active in your project.

Installation & Upgrades

Download & Import
After purchasing the package on the Unity Asset Store, open the **Package Manager** and download it from **My Assets**.
The default import provides the **URP** setup out of the box.

Upgrading to HDRP
If your project uses HDRP, open the included **Upgrade to HDRP** folder and follow the provided upgrade steps to switch the shader and materials to HDRP-compatible versions.

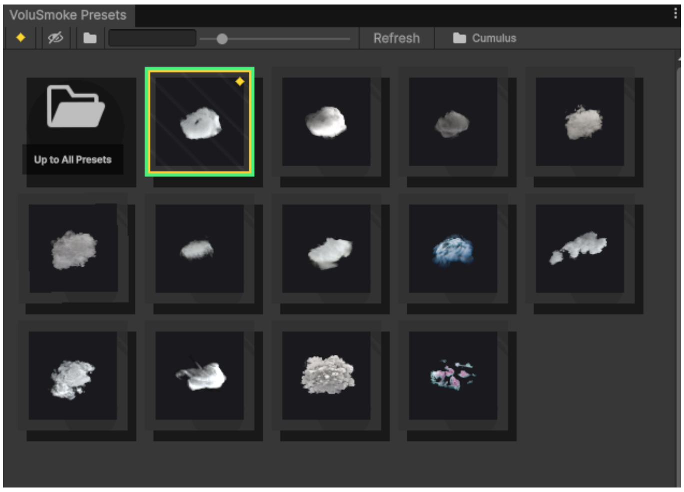
Creating Your First VoluSmoke Setup

In the Unity Hierarchy, right-click and navigate to:
VertexField → VoluSmokeFX → VoluSmoke Slice
This will spawn a VoluSmoke setup in your scene with a default preset applied.

Material

Material customization details will be added to this documentation in a future update. For now, you can learn about advanced material tweaking in the [Advanced Tutorial video](#).

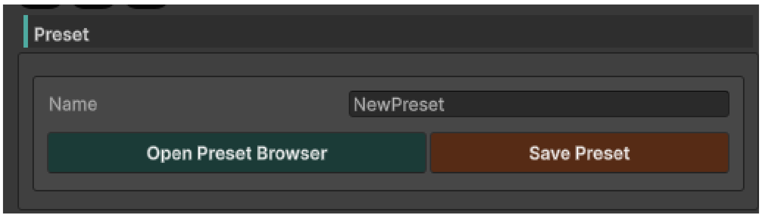
Preset Browser



VoluSmoke includes a built-in Preset Browser that lets you quickly browse, apply, and manage smoke and fog presets. You can instantly test different looks, save your own custom presets, organize them into folders, and mark favorites for faster access. This makes it easy to maintain a smooth and efficient workflow across multiple scenes and projects.

Open Preset Browser.

To access the Preset Browser, simply click the **"Open Preset Browser"** button. This will open a separate window displaying all available presets for easy browsing and selection.



Saving your presets.

To save a custom preset, enter a name for it and click **Save**. The new preset will appear in the **root (top) folder**, and you can move it into any other folder for organization.
If the Preset Browser window is already open while saving, the new preset may not appear immediately. Click 

to display the latest saved presets.button to see your latest saved preset.

Appling preset.

To apply a preset, simply select your **VoluSmoke Slice** in the Hierarchy, then click on any preset in the Preset Browser. The selected preset will be applied instantly.

Note: A VoluSmoke Slice must be selected before applying a preset — otherwise, the Preset Browser won't know which object to update.

Marking as Favorite.

To mark a preset as a favorite, right-click on it and select "Mark as Favorite". Favorited presets are highlighted with a gold border, making them easier to spot.

You can also toggle the ***Favorites Filter***  to show only your favourite presets in the browser for quick access.

Folders.

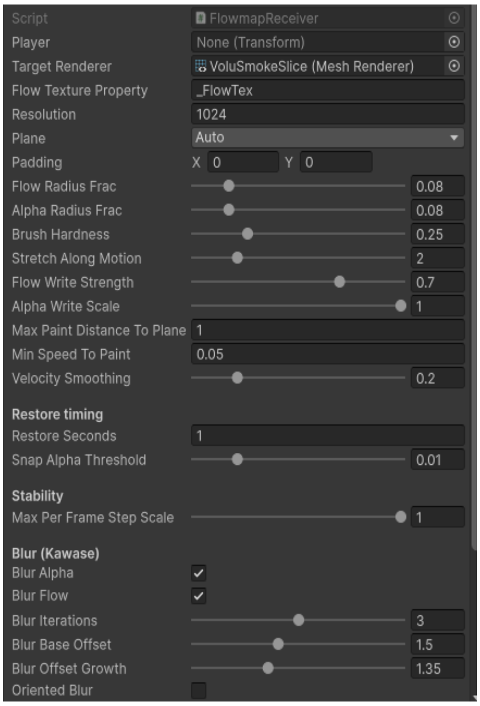
You can organize your presets by moving them into folders. To do this, right-click a preset and choose ***"Move to Folder"***, then select the target folder.
You can also toggle the ***Folder View*** icon to show or hide presets that are not assigned to any folder.

To create a new folder, click the Folder icon and select "***Add Folder…***". Alternatively, you can create a standard folder directly in: `Assets/VertexField/VoluSmoke FX/Presets/`
Any new folders added there will appear in the Preset Browser.

Starting Preset.

When you create your first VoluSmoke Slice, a default starting preset is automatically applied. This preset is highlighted with a green border in the Preset Browser.
If you want to change which preset is used as the default starting point, locate the preset file in the Presets folder, select it, and click "***Set as Starting Preset***". The newly assigned preset will then be used for all future newly created VoluSmoke Slices.

Interaction



To enable interaction with VoluSmoke, add the FlowmapReceiver.cs script to your VoluSmoke object.

Set the "Player" field to the object that should interact with the smoke, and adjust the script's settings to fine-tune the response.

This script is provided as an example implementation.

If you prefer to create your own interaction system, the only requirement is to supply a flowmap Render Texture to the VoluSmoke material. The Blue (B) channel of the flowmap controls the alpha used for the interaction effect.You can explore the Interaction Demo Scenes included in the package to see this feature in action and better understand how it works.

Support & Contact

- **Email Support:** vertexfield@gmail.com
- **Discord Community:** https://discord.gg/HnQ48uMN
- **Response Time:** We aim to reply to general inquiries within **2 business days**.

Bug Reporting Guidelines

When reporting an issue, please include the following to help us reproduce and resolve it quickly:

1. **Project Setup Details**
   Unity version, active render pipeline (URP/HDRP + version), and target platform/hardware.
2. **Reproduction Steps**
   A short description of how to reproduce the issue, ideally with a small test scene or project that demonstrates the problem.
3. **Logs & Output**
   Attach the `Editor.log` and/or `Player.log` files from the affected session.

Licensing

- This asset is licensed under the **standard Unity Asset Store terms**.
- You may use it in commercial and non-commercial projects, but **redistribution, reselling, or sharing of the source files, shaders, or assets outside of a compiled game or application is not permitted**.